

MEDICAL INVENTORY MANAGEMENT PLATFORM

Project Understanding and Objectives :

Project Understanding :

The Medical Inventory Management Platform is a full-stack web application developed using **React.js** for the frontend, **Spring Boot** for the backend, and **PostgreSQL** as the database. The application is designed for pharmacies, hospitals, and healthcare organizations to efficiently manage medicine inventory, monitor stock availability, track medicine expiry dates, manage supplier information, and generate inventory analytics.

The system provides secure user authentication using **JWT** and **OAuth2**, ensuring that only authorized users can access the application based on their roles. It also offers real-time stock updates, low-stock alerts, expiry notifications, supplier management, purchase order management, medicine search and filtering, and inventory dashboards to improve inventory control and reduce medicine wastage.

Project Objectives :

- Develop a secure and scalable web-based medicine inventory management system.
- Implement JWT and OAuth2 authentication for secure login.
- Manage medicine inventory with real-time stock updates.
- Monitor medicine expiry dates and generate expiry notifications.
- Detect low-stock medicines and notify users.
- Manage supplier records and purchase orders.
- Generate inventory analytics and reports.
- Provide role-based access control for different users.
- Deploy the application using React.js, Spring Boot, and PostgreSQL.

Problem Statement :

Many pharmacies, hospitals, and healthcare organizations still manage medicine inventory manually using spreadsheets or paper records. These traditional methods often result in inaccurate inventory records, medicine shortages, expired medicines remaining in stock, inefficient supplier management, and delayed reporting.

The Medical Inventory Management Platform automates the entire inventory management process by maintaining accurate medicine records, tracking stock availability, monitoring expiry dates, managing suppliers and purchase orders, and generating inventory reports. The system improves operational efficiency, reduces medicine wastage, and supports better decision-making through real-time inventory analytics.

User Roles And Permissions :

Admin

Permissions :

- Manage users
- Manage medicines
- Manage suppliers
- Manage inventory
- Manage purchase orders
- View dashboard
- Generate reports
- Manage notifications

Pharmacist

Permissions :

- Search medicines by name, category, or batch number
- View medicine details and stock availability
- Dispense medicines and update inventory
- Monitor expiry dates before dispensing medicines
- View low-stock alerts
- View near-expiry and expired medicines
- View supplier information for medicines
- Access inventory dashboard
- Generate medicine availability reports

Staff

Permissions :

- Add new medicine records
- Update medicine information
- View medicine details
- Add incoming stock
- Update stock quantities
- Record stock movement
- Manage supplier information
- Create purchase orders
- Receive purchased medicines into inventory
- Search and filter medicines
- View low-stock notifications
- View medicines approaching expiry
- Access inventory dashboard
- Generate inventory and purchase reports

Functional Requirements :

Authentication Module :

- User Registration
- User Login
- JWT Authentication
- OAuth2 Login
- Logout
- Password Encryption
- Role-Based Authorization

User Management Module

Medicine Management Module :

- Add Medicine
- Update Medicine
- Delete Medicine
- Search Medicine
- Filter Medicines by Category
- View Medicine Details

Medicine Details include:

- Medicine Name
- Category
- Manufacturer
- Batch Number
- Manufacturing Date
- Expiry Date
- Price
- Quantity
- Supplier

Inventory Management Module :

- Add New Stock
- Update Stock
- Reduce Stock
- Track Inventory
- View Stock History
- Monitor Stock Availability

Supplier Management Module :

- Add Supplier

- Update Supplier
- Delete Supplier
- View Supplier Information
- Manage Supplier Records

Purchase Management Module :

- Create Purchase Orders
- Update Purchase Orders
- View Purchase History
- Receive Purchased Medicines

Expiring Monitoring Module :

- Store Expiry Dates
- Identify Near Expiry Medicines
- Identify Expired Medicines
- Generate Expiry Reports

Notification Module :

- Low Stock Alerts
- Expiry Alerts
- Purchase Notifications Dashboard Notifications

Dashboard Module :

Display :

- Total Medicines
- Total Categories
- Total Suppliers
- Available Stock
- Low Stock Medicines
- Expired Medicines
- Near Expiry Medicines

- Purchase Orders
- Recent Activities

Report Modules :

Generate :

- Inventory Report
- Supplier Report
- Purchase Report
- Expiry Report
- Low Stock Report

Non-Functional Requirements

Security :

- JWT Authentication
- OAuth2 Authentication
- Password Encryption
- Secure REST APIs
- Role-Based Access Control

Performance :

- Fast response time
- Efficient database queries
- Optimized REST APIs

Reliability :

- Accurate inventory updates
- Data consistency
- Regular database backup

Scalability :

- Support thousands of medicines
- Support multiple concurrent users

Availability :

- Accessible through web browsers
- High system availability

Usability :

- Responsive user interface
- Easy navigation
- User-friendly dashboard

Maintainability :

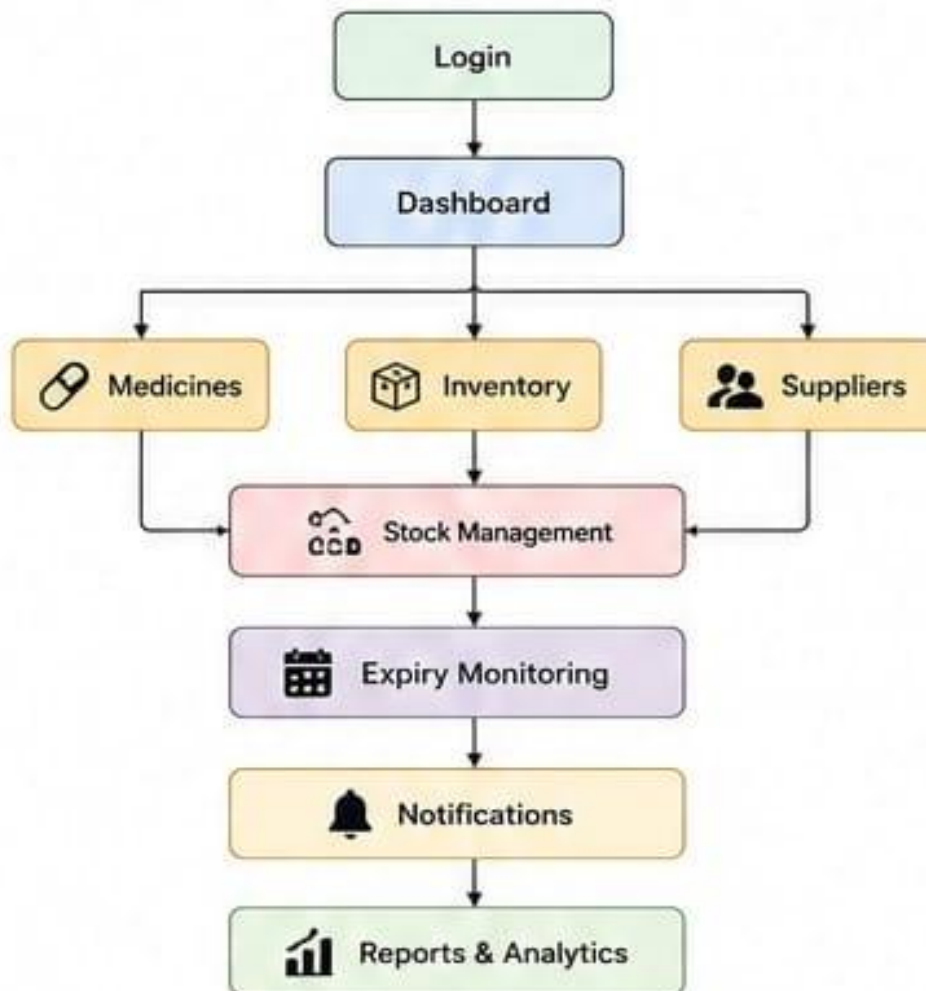
- Modular application architecture
- Reusable React components
- Layered Spring Boot architecture
- Well-documented code

Compatibility :

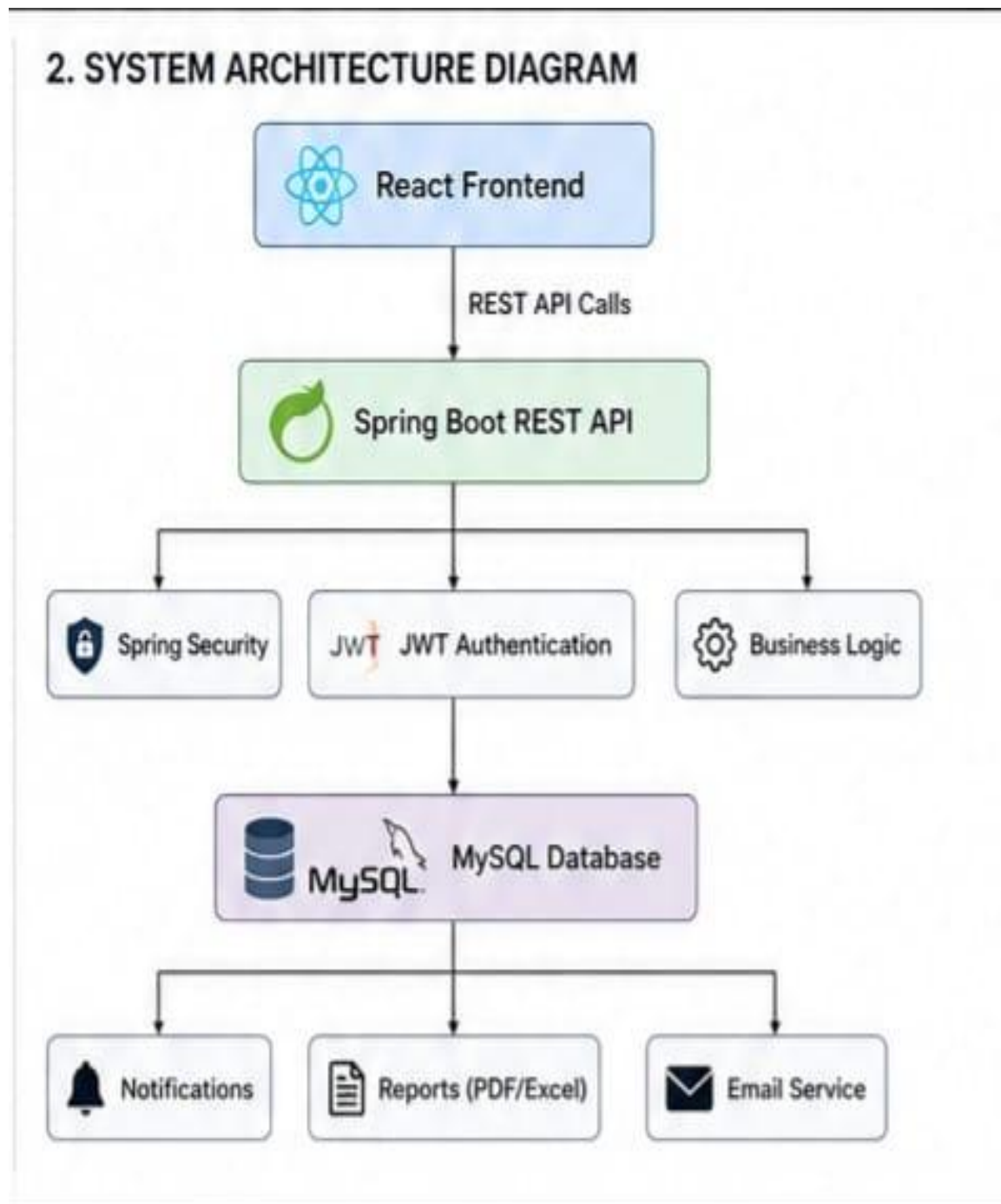
- Google Chrome
- Mozilla Firefox
- Microsoft Edge
- Mobile Responsive Design

Workflow Diagram :

1. WORKFLOW DIAGRAM

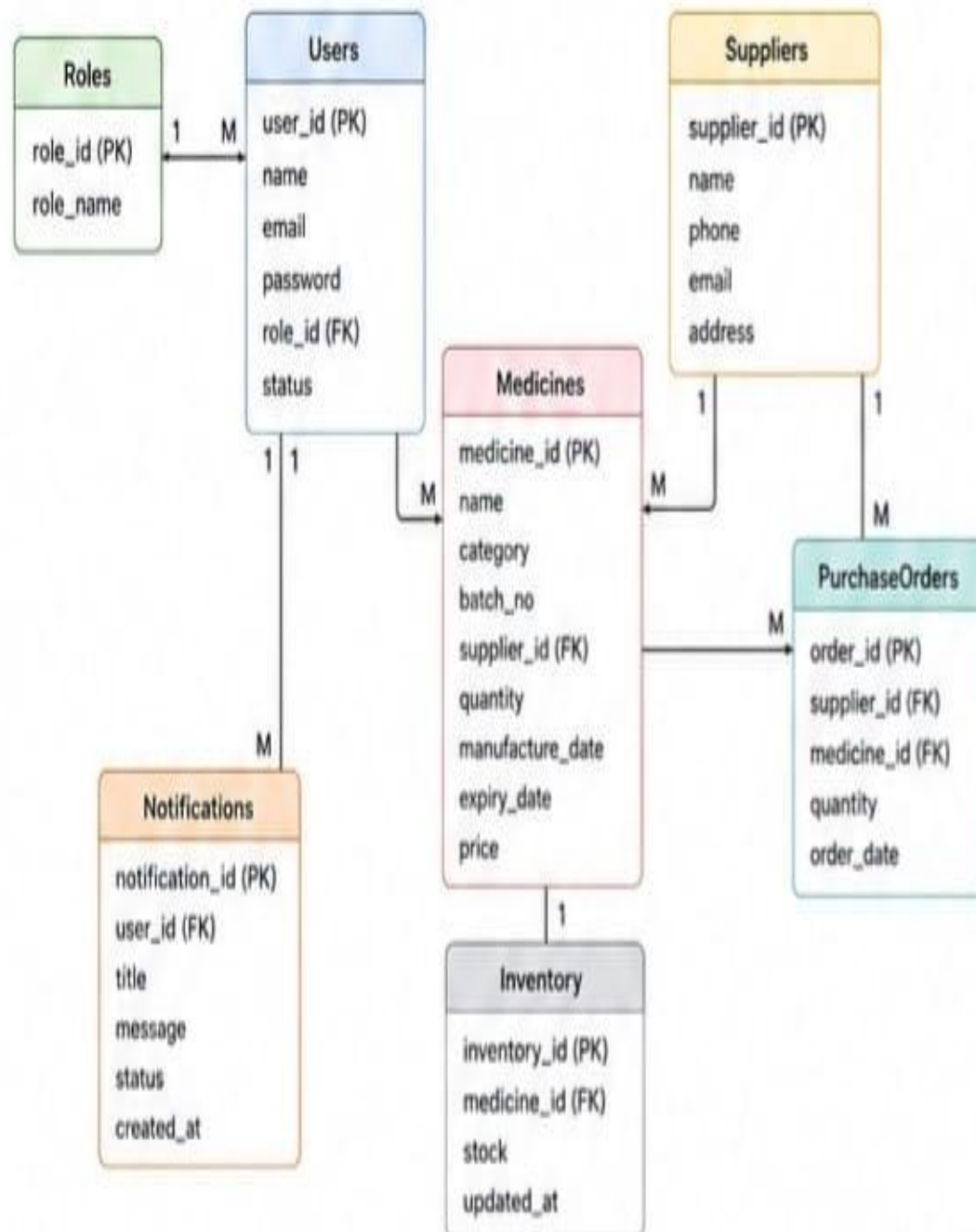


System Architecture Diagram :



Database Schema(ER Diagram) :

3. ER DIAGRAM (DATABASE SCHEMA)



UI Wireframes :

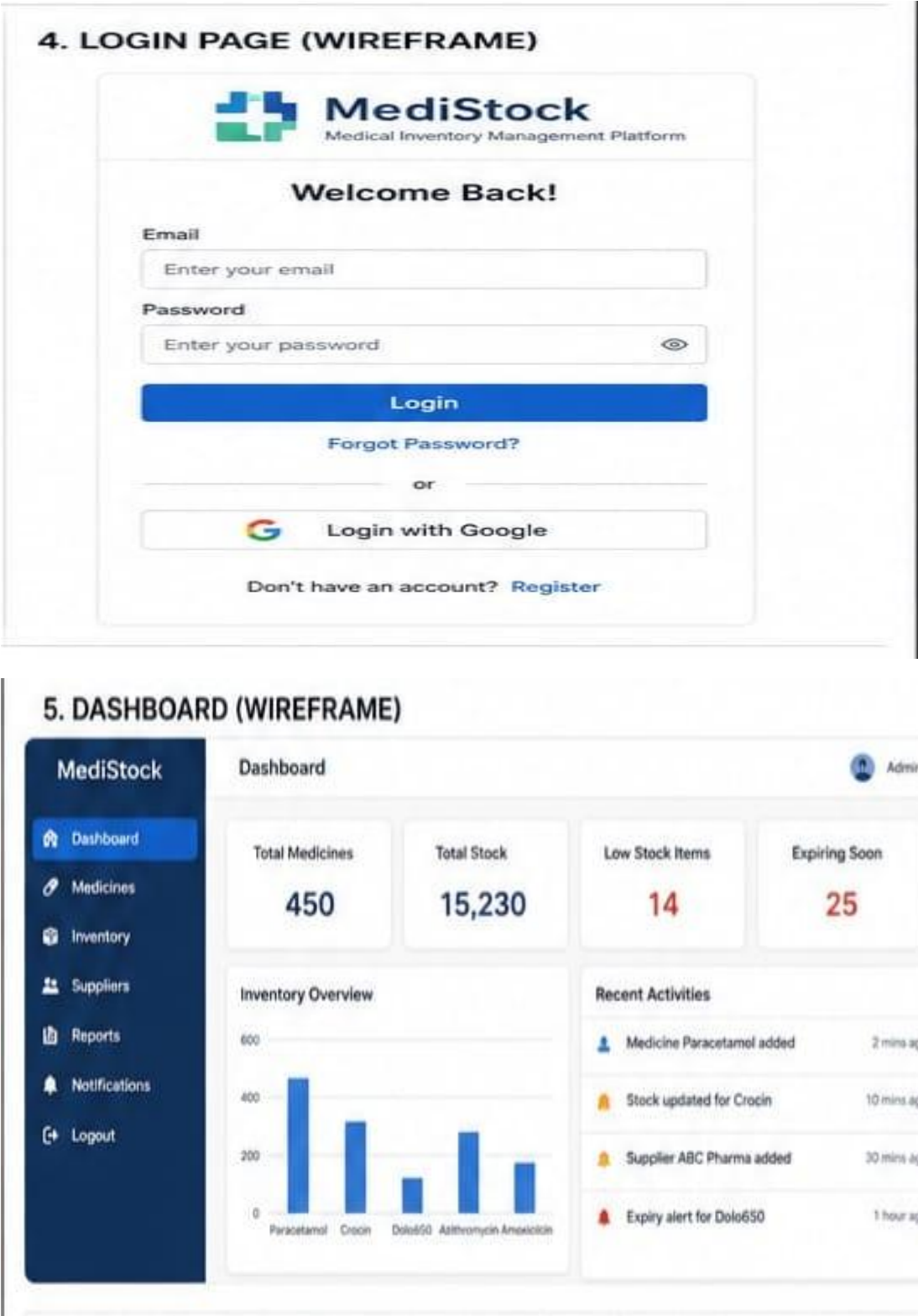


Fig : Login Page and Dashboard Page

6. ADD MEDICINE PAGE (WIREFRAME)

Add New Medicine

Medicine Name Enter medicine name	Category Select category
Batch Number Enter batch number	Quantity Enter quantity
Supplier Select supplier	Manufacturing Date dd-mm-yyyy
Price Enter price	Expiry Date dd-mm-yyyy

SaveCancel

Fig : Add Medicine Page

7. INVENTORY PAGE (WIREFRAME)

Filter

Medicine	Batch No	Stock	Expiry Date	Status	Action
Paracetamol	B1234	120	12/12/2027	Available	
Crocin	C2345	15	09/09/2026	Low Stock	
Dolo650	D4556	0	06/06/2026	Out of Stock	
Azithromycin	A6789	200	05/11/2027	Available	

Fig : Inventory Page

8. SUPPLIER PAGE (WIREFRAME)

Add Supplier

Name	Contact	Email	Action
ABC Pharma	9876543210	abc@gmail.com	 
XYZ Medical	9876543211	xyz@gmail.com	 
HealthCare Ltd	9876543212	health@gmail.com	 

Fig : Supplier Page

9. REPORTS PAGE (WIREFRAME)



Inventory Report

View and download inventory report



PDF



Excel



Purchase Report

View and download purchase report



PDF



Excel



Expiry Report

View and download expiry report



PDF



Excel



Supplier Report

View and download supplier report



PDF



Excel

Fig : Reports Page